

1. Problem

- Telecom BPO teams manually read every inbox item - slow, inconsistent, judgment-heavy.
- Billing, outages, escalations, SIM/port, plan changes, and FAQs need different remediation paths.
- SLA leakage and rework rise when routing depends on individual agent interpretation.
- Ops leaders need auditable proof that AI can classify AND execute branch-specific actions.

2. Solution - PulseDesk

- AI intake console: classify -> branch playbook -> remediate -> log (Streamlit + LLM/heuristic + SQLite).
- Structured classification: type, urgency, sentiment, entities, confidence.
- Governance gate: confidence < 65% parks cases On hold / Needs Review before unsupervised release.
- Agent HITL (edit / release / escalate) and Tech Lead queue (acknowledge / approve / return).
- Simulated tickets, holds, notifications, and follow-ups with a durable case audit trail.

3. Workflow design (6 branches)

- Billing dispute: provisional hold -> Billing ticket -> 48h follow-up -> acknowledgment draft.
- Service outage: bulletin match -> SLA/ETA reply OR Network Ops ticket if unmatched.
- Complaint / escalation: P1 Retention case -> Team Lead notify -> same-day callback -> recovery draft.
- SIM / port: identity checklist -> Port/SIM ticket -> 24h status follow-up.
- Plan change: eligibility -> catalog quote -> Care order -> confirmation follow-up.
- General enquiry: FAQ / self-serve -> light ticket -> close-or-route disposition.
- Every branch emits response + route + follow-up + Case Log persistence.

4. Demo outcomes

- Six golden samples (REQ-001 ... REQ-006) each trigger a distinct remediation timeline.
- Panel path: REQ-001 billing, REQ-002 outage, REQ-003 escalation - three branches under 3 minutes.
- Edge samples show the confidence gate (Needs Review / escalate to lead).
- Case Log proves classification, actions, and drafts are persisted for audit.
- Illustrative KPI: manual ~12 min first action -> assisted ~90 seconds (POC estimate).
- Live demo: <https://pulsedesk-firstsource.streamlit.app>
- GitHub: <https://github.com/samkrish13/FirstSourcePOC>
- Backup: <=3 min screen recording if Cloud is unavailable (PulseDesk_Demo_Script.md).

5. Scale path

- Connect real email / CRM / ITSM (ServiceNow, Salesforce) via production orchestration.
- Evaluation harness: labeled inbox set, precision/recall by branch, drift monitors.
- Human-in-the-loop for P1 and low-confidence; four-eyes on goodwill credits.
- Expand to chat / voice transcripts with the same remediation contracts.
- Hardening: OAuth mailbox, outbound send on release, role-based SSO.